LOG COLLECTION & SIEM ARCHITECTURE

Network Security Architecture Programme

Document Reference: SEC-DOC-004 | Version 1.0 | March 2026

CONFIDENTIAL — INTERNAL USE ONLY

SIEM: Wazuh | Log Collector: 172.16.66.20 | SIEM: 172.16.66.10 | SOC Dashboard: 172.16.66.30

1. Document Information

Document Title	Log Collection & SIEM Architecture
Document Reference	SEC-DOC-004
Version	1.0
Date	March 2026
Classification	**CONFIDENTIAL — INTERNAL USE ONLY**
Status	**APPROVED — Log collection paths confirmed from architecture review. Wazuh SIEM operational.**
SIEM Platform	Wazuh Open-source XDR / SIEM
Log Collector IP	172.16.66.20
Wazuh SIEM IP	172.16.66.10
SOC Dashboard IP	172.16.66.30
SOC Zone VLAN	VLAN 40 Security/SOC — 10.10.40.0/24 (user VLAN) | SOC servers reside on 172.16.66.0/24
Related Documents	SEC-DOC-001 Architecture Overview v2.0 | SEC-DOC-003 Firewall Policy | SEC-DOC-008 Controls Checklist v2.0

1.1 Purpose

This document describes the log collection architecture, Wazuh SIEM platform configuration, detection rules, alert procedures, log retention policy, and operational processes for the organisation's Security Operations Centre. It provides the authoritative reference for how security events are collected from every zone and device, correlated in the SIEM, and actioned by SOC analysts.

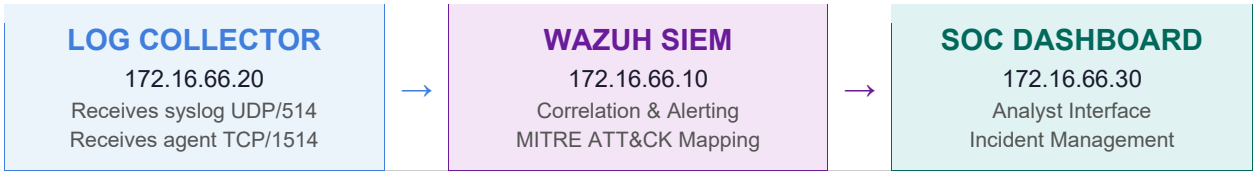
1.2 Scope

All 14 security zones, all 16 syslog-capable network devices, all 23 Wazuh-agent-monitored servers, the Log Collector, Wazuh SIEM, and SOC Dashboard. Includes NTP synchronisation, alert rules, MITRE ATT&CK mapping, log retention, log integrity, and SOC operating procedures.

2. SOC Zone Architecture

2.1 SOC Pipeline Overview

Three components form the centralised monitoring pipeline. All log data flows left-to-right through this chain: sources send to the Log Collector, which normalises and forwards to Wazuh SIEM, which drives the SOC Dashboard for analyst interaction.



2.2 Log Collector (172.16.66.20)

Single ingestion point for all log data. Receives syslog (UDP/514) from all network devices and Wazuh agent connections (TCP/1514) from all servers. Normalises timestamps against NTP, buffers 24 hours of events locally, and forwards to Wazuh SIEM. No log data bypasses the Log Collector.

2.3 Wazuh SIEM (172.16.66.10)

Open-source XDR and SIEM platform. Provides real-time event correlation, MITRE ATT&CK-mapped rule-based alerting, file integrity monitoring via agents, vulnerability detection against the NVD feed, and compliance dashboards for PCI-DSS, ISO 27001, and CIS Controls. SIEM configuration and rule management are performed by the SOC Lead.

2.4 SOC Dashboard (172.16.66.30)

Analyst-facing interface. Access restricted to authorised SOC staff via AD-enforced RBAC. Provides real-time alert queue, threat hunting query interface, pre-built compliance dashboards, certificate expiry monitoring, and report generation for audit and management purposes. Dashboard access is read-only for analysts — no ability to delete or modify event records.

3. Log Source Inventory

3.1 Network Device Log Sources — Syslog (UDP/514)

All network devices forward syslog events at severity level Notice (5) and above to the Log Collector. No agent required.

Device	IP Address	Zone	Method	Proto	Port	Status
Perimeter Firewall	203.0.113.4 / 172.16.10.1	Perimeter	Syslog	UDP	514	**ACTIVE**
Internal Firewall	172.16.21.10	Internal FW	Syslog	UDP	514	**ACTIVE**
DC Firewall	172.16.30.1	Data Centre	Syslog	UDP	514	**ACTIVE**
DB Firewall	172.16.30.x	Data Centre	Syslog	UDP	514	**ACTIVE**
PCI Firewall	172.16.50.x	PCI / ATM	Syslog	UDP	514	**ACTIVE**
Branch Firewall	10.12.x.x	Branch	Syslog	UDP	514	**ACTIVE**
Core Router 1	10.0.0.1	Core Net	Syslog	UDP	514	**ACTIVE**
Core Router 2	10.0.0.2	Core Net	Syslog	UDP	514	**ACTIVE**
Core Switch 1	10.0.1.1	Core Net	Syslog	UDP	514	**ACTIVE**
Core Switch 2	10.0.1.2	Core Net	Syslog	UDP	514	**ACTIVE**
WAF	172.16.10.10	DMZ	Syslog	UDP	514	**ACTIVE**
Reverse Proxy	172.16.10.11	DMZ	Syslog	UDP	514	**ACTIVE**
Load Balancer	172.16.10.12	DMZ	Syslog	UDP	514	**ACTIVE**
Email Gateway	172.16.10.41	DMZ	Syslog	UDP	514	**ACTIVE**
ATM Gateway	Dedicated	ATM Network	Syslog	UDP	514	**ACTIVE**
Branch Router	10.12.x.x	Branch	Syslog	UDP	514	**ACTIVE**

3.2 Server Log Sources — Wazuh Agent (TCP/1514)

All servers have the Wazuh agent installed. Agents connect outbound to the Log Collector on TCP/1514 and provide OS log collection, application log collection, FIM, and vulnerability detection.

Server	IP Address	Zone	Method	Port	Status
Banking App	172.16.20.10	Application Services	Wazuh Agent	1514	**ACTIVE**
Mobile Backend	172.16.20.11	Application Services	Wazuh Agent	1514	**ACTIVE**
IAM Server	172.16.20.12	Application Services	Wazuh Agent	1514	**ACTIVE**
MFA Server	172.16.20.13	Application Services	Wazuh Agent	1514	**ACTIVE**
Active Directory	172.16.60.10	Identity Services	Wazuh Agent	1514	**ACTIVE**

Server	IP Address	Zone	Method	Port	Status
DNS (Internal)	172.16.60.11	Identity Services	Wazuh Agent	1514	**ACTIVE**
RADIUS Server	172.16.60.12	Identity Services	Wazuh Agent	1514	**ACTIVE**
DHCP Server	172.16.60.13	Identity Services	Wazuh Agent	1514	**ACTIVE**
Core Banking Server	172.16.30.x	Data Centre	Wazuh Agent	1514	**ACTIVE**
Core Banking DB	172.16.30.20	Data Centre	Wazuh Agent	1514	**ACTIVE**
Transaction DB	172.16.30.21	Data Centre	Wazuh Agent	1514	**ACTIVE**
Backup Server	172.16.30.40	Data Centre	Wazuh Agent	1514	**ACTIVE**
Payment Gateway	172.16.50.10	Payment / PCI	Wazuh Agent	1514	**ACTIVE**
Card Processing	172.16.50.20	Payment / PCI	Wazuh Agent	1514	**ACTIVE**
Transaction Server	172.16.50.30	Payment / PCI	Wazuh Agent	1514	**ACTIVE**
Web Server	172.16.10.20	DMZ Web Servers	Wazuh Agent	1514	**ACTIVE**
API Gateway	172.16.10.30	DMZ Web Servers	Wazuh Agent	1514	**ACTIVE**
Public DNS	172.16.10.40	DMZ	Wazuh Agent	1514	**ACTIVE**
Log Collector (self)	172.16.66.20	SOC	Local agent	—	**ACTIVE**
Wazuh SIEM (self)	172.16.66.10	SOC	Self-monitor	—	**ACTIVE**
DR Replication Node	TBD	DR Site	Wazuh Agent	1514	**PENDING**
DR Database	TBD	DR Site	Wazuh Agent	1514	**PENDING**
DR Backup	TBD	DR Site	Wazuh Agent	1514	**PENDING**

DR Agents PENDING: Wazuh agents on DR nodes cannot be deployed until IP addresses are assigned (Observation DR-08, SEC-DOC-008 v2.0). SOC team to deploy agents as part of DR provisioning — Q2 2026.

4. Log Types Collected per Source

4.1 Network Device Log Types

Device	Zone	Log Types Collected
All firewalls (FW-01–06)	Multiple	Allow/deny events, session logs, management access attempts, config changes, interface state changes, NAT translation events
Core Routers (10.0.0.1/2)	Core Network	BGP/OSPF state changes, routing table events, CPU/memory alerts, ACL hits, interface up/down events
Core Switches (10.0.1.1/2)	Core Network	STP topology changes, port security violations, DHCP snooping events, MAC address table changes, VLAN events
WAF (172.16.10.10)	DMZ	OWASP rule hits, blocked and detected requests, SQLi/XSS/LFI/SSRF attack patterns, rate limit events, TLS errors
Reverse Proxy (172.16.10.11)	DMZ	HTTP access logs, error logs, backend connection failures, SSL certificate events, upstream response codes
Load Balancer (172.16.10.12)	DMZ	Connection distribution events, backend health check failures, session persistence events
Email Gateway (172.16.10.41)	DMZ	Inbound/outbound mail events, spam and malware detections, quarantine events, DMARC/SPF/DKIM results
ATM Gateway	ATM Network	ATM transaction events, connection state changes, error codes, firmware events, tamper detection alerts
Branch Firewall/Router	Branch Office	VPN tunnel state changes, IKEv2 negotiation events, ACL hits, authentication events

4.2 Server Log Types — Wazuh Agent

Server Group	Log / Data Types	FIM Paths
Banking App & Mobile Backend	Application logs, authentication events, session creation/termination, API access logs, error and exception logs	App binary dirs, /etc/app-config, web root, SSL cert paths
IAM Server & MFA Server	Auth success/failure, token issuance/revocation, policy changes, admin access events, session logs	/etc/iam/, /etc/mfa/, config and policy directories
Active Directory	Windows Security Event Log — events 4624,4625,4648,4720,4722,4740,4768,4769,4776,4798,4799 — Group Policy changes, AD object modifications	System32, SysWOW64, SYSVOL, NTDS directory
Identity Services (DNS, RADIUS, DHCP)	DNS query logs (all queries), zone transfer attempts, RADIUS auth events, DHCP lease grants and renewals, config changes	Service binary and config directories
Core Banking Server	Application and OS logs, DB connection events, transaction processing logs, startup/shutdown events	Application directories, /etc, DB connection config
Core Banking DB & Transaction DB	Database audit logs (all statements on sensitive tables), login events, failed auth, schema changes, privilege grants, export operations	DB data directory, config files, audit log paths

Server Group	Log / Data Types	FIM Paths
Payment Processing (all 3 servers)	All PCI transaction logs, cardholder data access events, config changes, admin access, process events — PCI-DSS Req 10 mandatory	ALL files on PCI-scope servers — CRITICAL FIM alert on any change
DMZ Web & API Servers	HTTP access logs, error logs, API call logs, TLS handshake events, application exceptions	Web content dirs, config files, SSL cert paths

5. NTP Synchronisation

Accurate timestamps are critical for event correlation. A time discrepancy of more than 1 second between log sources can break SIEM correlation chains. All devices are synchronised to a Stratum 2 internal NTP server. Stratum 2 sync is confirmed for all in-scope devices.

Stratum	NTP Source	Devices Synchronising
Stratum 1	External: pool.ntp.org	Internal NTP Server synchronises to external Stratum 1 source
Stratum 2	Internal NTP Server (172.16.66.x)	All firewalls, routers, switches, servers, Wazuh SIEM, Log Collector, SOC Dashboard — primary NTP source for all in-scope devices. CONFIRMED.
Stratum 3 (fallback)	Secondary NTP (Identity Services zone)	Fallback NTP if primary unavailable. All devices configured with both primary and secondary NTP server addresses.

> **SIEM Alert:** NTP drift > 1 second triggers MEDIUM severity alert. Drift > 5 seconds triggers HIGH severity alert. Either may indicate misconfiguration or a time-manipulation attack aimed at disrupting log correlation.

6. Wazuh SIEM Alert Rules & Detection Coverage

6.1 Alert Severity Levels

Wazuh Level	Severity	SOC Response SLA	Examples
15	CRITICAL	Immediate — auto-page on-call analyst. Auto-create incident.	DB tier access violation, non-OOB management access, PCI zone implicit deny
12–14	HIGH	15 minutes — on-call analyst investigates.	Brute force success, privilege escalation, FIM on critical binary, FW config change outside change window
8–11	MEDIUM	1 hour — analyst reviews next shift.	Multiple login failures, OWASP rule hit, new service installed, unusual outbound volume
4–7	LOW	8 hours — reviewed in daily SOC report.	Single failed login, standard deny events, low-confidence anomaly
1–3	INFO	Weekly batch review.	Informational events, NTP sync, standard connection events

6.2 Detection Rule Categories

Rule Category	MITRE Tactic	Severity	Log Sources	Detection Logic
Authentication Brute Force	Credential Access	HIGH	AD, IAM, RADIUS, SSH	5+ failed logins within 60 seconds from same source IP
Login Success After Failures	Initial Access	MEDIUM	AD, IAM, MFA logs	Successful login following 3+ failures in prior 10 minutes
Privilege Escalation	Privilege Escalation	HIGH	AD Event Log, sudo logs	New admin group membership, sudo/UAC bypass, token manipulation
Pass-the-Hash / Kerberoast	Lateral Movement	HIGH	AD Event 4768/4769/4776	NTLM auth in Kerberos env, RC4 downgrade, AS-REP roasting pattern
Firewall Config Change	Defence Evasion	HIGH	All FW syslog	Config change event outside approved maintenance window
DB Tier Access Violation	Discovery / Intrusion	CRITICAL	DB Firewall syslog	Any connection hitting DBF-008 implicit deny on database tier
PCI Zone Access Violation	Exfiltration	CRITICAL	PCI Firewall syslog	Unauthorised access attempt to payment zone (PCF-023 triggered)
FIM — Critical Binary Modified	Tampering	HIGH	Wazuh FIM	OS binary, application executable, or monitored config file modified
FIM — PCI Scope Event	Tampering	CRITICAL	Wazuh FIM (PCI servers)	Any FIM event on PCI servers — cannot be suppressed. PCI-DSS Req 11.5.
Malware / Suspicious Process	Execution	HIGH	Wazuh agent process	Known malware hash, suspicious parent/child

Rule Category	MITRE Tactic	Severity	Log Sources	Detection Logic
				process, encoded command strings
DNS Tunnelling / C2 Pattern	C2 / Exfiltration	**MEDIUM**	DNS server (172.16.60.11)	High-entropy subdomain, beaconing pattern, query length anomaly, known malicious domain
OWASP Attack — WAF Block	Initial Access	**MEDIUM**	WAF syslog	SQLi, XSS, LFI, SSRF, command injection blocked by WAF
Unusual Outbound Volume	Exfiltration	**HIGH**	All FW syslog	Outbound data volume from server significantly above daily baseline
VPN Tunnel State Change	Impact	**MEDIUM**	Branch FW/Router syslog	IPSec tunnel drop, failed re-establishment, unexpected peer IP
Non-OOB Management Attempt	Defence Evasion	**CRITICAL**	All FW management syslog	Any attempt to access any firewall management from non-OOB source
Certificate Expiry Warning	Impact	**MEDIUM**	SOC Dashboard monitoring	Internet-facing certificate within 30 days of expiry
Log Source Silence	Defence Evasion	**HIGH**	Wazuh SIEM internal	No events received from agent or syslog source in last 5 minutes
NTP Drift Exceeded	Tampering	**MEDIUM**	Wazuh SIEM internal	Device NTP drift exceeds 1-second threshold vs internal NTP server
New User Account Created	Persistence	**MEDIUM**	AD Security Event 4720	User account created — verify against approved HR/change request
Service Account Interactive Login	Anomalous Activity	**HIGH**	AD Security Event 4624	Service account (svc_*) logged in interactively from any workstation
Replication Lag — RPO Breach	Impact	**HIGH**	DR replication monitoring	DB replication lag to DR site exceeds 15-minute RPO threshold
DHCP Snooping Violation	Lateral Movement	**MEDIUM**	Core Switch syslog	Rogue DHCP server detected on untrusted port
STP Topology Change	Lateral Movement	**MEDIUM**	Core Switch syslog	Unexpected STP topology change — possible network bridge insertion

7. MITRE ATT&CK Coverage

Current MITRE ATT&CK coverage is estimated at 60% of relevant techniques. Target is 80% by Q3 2026 via custom rule development and expanded threat intelligence integration.

Tactic	Coverage	% Covered	Priority Rules Active
Initial Access	4 of 6	**67%**	WAF OWASP blocks, VPN brute force, email attachment detection, external exploit patterns
Execution	3 of 5	**60%**	Suspicious process, PowerShell encoded commands, malicious script detection
Persistence	3 of 6	**50%**	New user account, new service, scheduled task creation, startup modification
Privilege Escalation	4 of 6	**67%**	Admin group membership change, sudo/UAC events, token impersonation, service privilege change
Defence Evasion	4 of 8	**50%**	FW config change outside window, log source silence, non-OOB mgmt access, FIM binary change
Credential Access	4 of 5	**80%**	Brute force, Pass-the-Hash, Kerberoasting indicators, LSASS access patterns
Discovery	2 of 5	**40%**	Port scan via FW deny spikes, AD enumeration events — partial coverage
Lateral Movement	3 of 4	**75%**	Pass-the-Hash, remote service auth anomalies, suspicious SMB/RDP from servers
Collection	2 of 4	**50%**	Unusual file access on sensitive paths, DB query volume anomaly
Exfiltration	3 of 5	**60%**	Unusual outbound volume, DNS tunnelling, PCI zone access violation
Impact	3 of 4	**75%**	Ransomware FIM pattern (mass file modification), service disruption, replication failure
Command & Control	2 of 5	**40%**	DNS C2 patterns, known C2 domain hits from threat intel — expanding Q2 2026

> **Q3 2026 Target:** Develop custom rules for Discovery and C2 tactics. Integrate two additional threat intel feeds (SC-10). Target overall MITRE ATT&CK coverage 80%+.

8. Log Retention Policy

PCI-DSS v4.0 Req 10.7 mandates 12 months online retention with 3 months immediately available. Current online retention of 6 months for non-PCI events is non-compliant. Extension to 12 months is targeted for Q2 2026 (SC-03 in SEC-DOC-008 v2.0).

Log Type	Online Retention	Archive Retention	Storage Location	Status
All SIEM events (normalised)	6 months (target: 12 months Q2 2026)	7 years	Wazuh SIEM + Archive	PARTIAL
PCI-scope events — all PCI zone logs	12 months	7 years (mandatory)	Dedicated PCI log storage + DR Backup	COMPLIANT
Firewall syslog (raw)	3 months	1 year	Log Collector + Archive	PARTIAL
AD Security Event Logs	12 months	7 years	SIEM + Archive	COMPLIANT
Wazuh FIM events	12 months	7 years	SIEM + Archive	COMPLIANT
Application logs (Banking, Mobile)	6 months	3 years	SIEM + Archive	PARTIAL
DB audit logs (Core Banking, Txn DB)	12 months	7 years	SIEM + Dedicated DB audit storage	COMPLIANT
VPN / Branch connection logs	3 months	1 year	SIEM + Archive	COMPLIANT
Email Gateway logs	6 months	3 years	Email GW + SIEM	COMPLIANT
SIEM alert records	Indefinite (compressed after 90 days)	Indefinite	SIEM + Archive	COMPLIANT

Action Required: Extend online retention from 6 months to 12 months for all non-PCI event types by Q2 2026 (SC-03). This is mandatory for PCI-DSS Req 10.7 compliance at next QSA assessment.

9. Log Integrity Protection

PCI-DSS v4.0 Req 10.5 requires log data to be protected from modification and unauthorised deletion. Current controls are partially compliant. WORM storage implementation is planned for Q2 2026 (SC-07 in SEC-DOC-008 v2.0).

9.1 Current Controls

- Log Collector receives logs via one-way syslog and agent forwarding — source devices cannot delete or modify events once forwarded
- Wazuh SIEM access restricted to SOC Lead and authorised analysts via AD RBAC — no direct database access for routine operations
- SOC Dashboard is read-only for analysts — no ability to delete or modify historical event records from the dashboard interface
- SIEM configuration changes require change management approval and are logged

9.2 Planned Controls (Q2 2026 — SC-07)

- WORM storage for all SIEM event indices — events immutable after ingestion
- Cryptographic log signing — each log batch timestamped and signed; hash stored independently from log data
- Nightly log integrity verification script — SIEM alerts if computed hash does not match stored events
- Immutable audit trail for all SIEM administrative actions

> **PCI Note:** PCI-DSS v4.0 Requirement 10.5.1 mandates log protection from modifications. WORM implementation is required for PCI compliance at next QSA review. PCI-scope logs on dedicated storage already have access controls in place — full WORM compliance by Q2 2026.

10. SOC Operating Procedures

10.1 Alert Triage Procedure

Step	Action	Owner	Time Target
1	Alert fires in SOC Dashboard — on-call analyst paged automatically for CRITICAL/HIGH	Wazuh SIEM auto-page	Automatic
2	Analyst acknowledges alert and assigns to self	On-call SOC Analyst	5 min (CRITICAL) / 15 min (HIGH)
3	Correlate with related events — pivot on source IP, username, or host	On-call SOC Analyst	Within 15 minutes of acknowledgement
4	Determine: True Positive / False Positive / Requires escalation	On-call SOC Analyst	Within 30 minutes
5a	False Positive: document reason, tune Wazuh rule, close ticket	On-call SOC Analyst	Within 1 hour
5b	True Positive Low/Medium: create incident ticket, assign to next shift, document all findings	On-call SOC Analyst	Within 1 hour
5c	True Positive High/Critical: invoke Incident Response. Notify SOC Lead and CISO immediately.	On-call Analyst + SOC Lead	Immediately
6	Document all steps, queries, pivots, and evidence in incident ticket throughout investigation	SOC Analyst	Ongoing throughout
7	Close ticket with resolution summary and any rule tuning actions taken	SOC Analyst	Before shift end

10.2 Proactive Threat Hunting Schedule

- Weekly: review all LOW events for slow brute force patterns and low-and-slow exfiltration not triggering alerts
- Weekly: review DNS query logs for high-entropy domains and server beaconing patterns
- Fortnightly: review AD logs for dormant account usage, admin account anomalies, and unusual logon hours
- Monthly: review all firewall implicit-deny logs for systematic internal subnet probing
- Monthly: confirm all Wazuh agents are active and reporting — validate against server inventory
- Quarterly: review MITRE ATT&CK coverage gaps and develop new detection rules for uncovered techniques

10.3 SIEM Tuning Policy

- Any rule generating more than 20 false positives per day must be reviewed within 5 business days
- Tuning changes documented with before/after rule comparison and business justification
- Rules may not be disabled entirely without SOC Lead approval — suppression preferred over disabling
- PCI-scope FIM rules and PCI zone access rules may not be tuned or suppressed without CISO approval — mandatory for PCI-DSS

11. Compliance Framework Mapping

Framework	Requirement	Control Area	How Satisfied
PCI-DSS v4.0	Req 10.2	Log all system component access	Wazuh agents on all PCI servers. AD Security Event Log forwarded. All access events captured.
PCI-DSS v4.0	Req 10.3	Protect audit logs	One-way log forwarding. RBAC on SIEM. WORM planned Q2 2026.
PCI-DSS v4.0	Req 10.4	Review logs daily	SOC Dashboard reviewed 24x7. CRITICAL/HIGH auto-paged immediately.
PCI-DSS v4.0	Req 10.5	Retain logs 12 months	PCI logs: 12 months online. Non-PCI: extending to 12 months Q2 2026.
PCI-DSS v4.0	Req 10.7	SIEM detecting anomalies	Wazuh SIEM deployed. 23 detection categories. MITRE mapping active.
PCI-DSS v4.0	Req 11.5	FIM on critical files	Wazuh FIM on all PCI servers. CRITICAL alert on any modification.
ISO 27001:2022	A.12.4.1	Event logging	Syslog from 16 network devices + Wazuh agents on 23 servers. All events centralised.
ISO 27001:2022	A.12.4.3	Admin and operator logs	OOB management access logged. AD admin events captured. SOC audit trail.
ISO 27001:2022	A.12.4.4	Clock synchronisation	NTP Stratum 2 confirmed on all devices. SIEM drift alert active.
CIS Controls v8	CIS 8.2	Collect audit logs	Centralised log collection from all sources via Log Collector.
CIS Controls v8	CIS 8.9	Centralise audit logs	All logs normalised and centralised in Wazuh SIEM (172.16.66.10).
NIST SP 800-53	AU-2	Event logging	All required event types captured across all 14 zones.
NIST SP 800-53	AU-9	Protection of audit info	RBAC on SIEM. WORM planned. One-way log forwarding in place.
NIST SP 800-53	AU-12	Audit record generation	Wazuh agents generate standardised audit records on all server types.

12. Open Items

ID	Item	Action	Owner	Target
SC-03	Online log retention is 6 months — PCI-DSS requires 12 months minimum	Extend to 12 months. Define 7-year archive policy.	SOC	**Q2 2026**
SC-05	MITRE ATT&CK coverage at 60% of techniques	Build custom rules for Discovery and C2 tactics	SOC	**Q3 2026**
SC-07	SIEM log integrity not enforced via WORM storage	Implement WORM storage. Implement cryptographic log signing.	SOC	**Q2 2026**
SC-10	Only one threat intel feed active	Integrate two additional commercial threat intel feeds	SOC	**Q2 2026**
DR-06	Wazuh agents not deployed on DR nodes (IPs not assigned)	Deploy agents once DR-08 (IP provisioning) is resolved	SOC + DC Ops	**Q3 2026**

Appendix A: Wazuh Agent Configuration Reference

Standard configuration applied to all Wazuh agents, managed centrally from Wazuh SIEM (172.16.66.10) and pushed to agents via the Wazuh manager.

Parameter	Value / Description
Wazuh Manager Connection Target	172.16.66.20 (Log Collector) — agents connect to Log Collector, not directly to SIEM
Agent Port	TCP/1514
Communication Encryption	AES-256 encrypted channel between agent and Log Collector
OS Log Collection — Linux	/var/log/auth.log, /var/log/syslog, /var/log/messages, auditd logs (/var/log/audit/audit.log)
OS Log Collection — Windows	Security, System, Application Event Logs — all events (no filtering at agent level)
Application Logs	Banking App / Mobile Backend: /var/log/app/ and application-defined paths DB logs: PostgreSQL/Oracle audit log paths PCI servers: all audit-scope log paths
FIM — All Servers (Linux)	/bin, /sbin, /usr/bin, /usr/sbin, /etc, /boot, /lib
FIM — All Servers (Windows)	C:\Windows\System32, C:\Windows\SysWOW64, C:\Program Files, C:\Program Files (x86)
FIM — Application Servers	Application binary directories and configuration file paths (defined per server role in Wazuh group config)
FIM — PCI Servers	ALL files on PCI-scope servers. CRITICAL Wazuh alert on any modification. Cannot be suppressed or tuned without CISO approval.
FIM Mode	Real-time inotify (Linux) / audit-based (Windows). Full FIM scan daily at 02:00.
Vulnerability Detection	Enabled. Pulls NVD feed daily at 03:00. Alerts on CVSS score >= 7.0 on detected vulnerabilities.

Parameter	Value / Description
Agent Keepalive	60-second heartbeat. Log source silence alert after 5 minutes of no events received.
Agent Auto-restart	Enabled — agent process auto-restarts on unexpected exit.
Active Response	Disabled globally except on WAF and Perimeter FW — IP block on CRITICAL brute force confirmed as true positive.

Appendix B: Revision History

Version	Date	Author	Description
1.0	March 2026	SOC Team / Security Architecture Team	Initial release. 16 syslog sources and 23 Wazuh agent sources documented. SOC three-component pipeline defined. NTP Stratum 2 confirmed. 23 detection rule categories documented. MITRE ATT&CK coverage assessed at 60%. Retention policy documented — SC-03 (6-month gap), SC-07 (WORM), SC-10 (threat intel), DR-06 (DR agents) tracked as open items.

Appendix C: Related Documents

Ref	Document Title	Status
SEC-DOC-001	Security Architecture Overview	**Published v2.0**
SEC-DOC-002	Network Segmentation & VLAN Design	Pending
SEC-DOC-003	Firewall Policy & Rules	**Published v1.0**
SEC-DOC-004	Log Collection & SIEM Architecture (this document)	**Published v1.0**
SEC-DOC-005	Identity & Access Management (IAM)	**Published v1.0**
SEC-DOC-006	Data Flow Diagram & Traffic Matrix	Pending
SEC-DOC-007	Disaster Recovery & Backup Plan	**Published v1.0**
SEC-DOC-008	Security Controls & Hardening Checklist	**Published v2.0**